

```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, confusion_matrix
```

```
In [3]: df = pd.read_csv("Social_Network_Ads.csv")
df
```

```
Out[3]:
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
...	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

400 rows × 5 columns

```
In [4]: df.isnull()
```

```
Out[4]:
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
395	False	False	False	False	False
396	False	False	False	False	False
397	False	False	False	False	False
398	False	False	False	False	False
399	False	False	False	False	False

400 rows × 5 columns

```
In [5]: df.isnull().sum()
```

```
Out[5]: User ID      0
Gender      0
Age         0
EstimatedSalary  0
Purchased   0
dtype: int64
```

```
In [6]: df.duplicated
```

```
Out[6]: <bound method DataFrame.duplicated of      User ID  Gender  Age  EstimatedSala
ry  Purchased
0    15624510   Male   19         19000      0
1    15810944   Male   35         20000      0
2    15668575  Female  26         43000      0
3    15603246  Female  27         57000      0
4    15804002   Male   19         76000      0
..      ...    ...    ...      ...      ...
395  15691863  Female  46         41000      1
396  15706071   Male   51         23000      1
397  15654296  Female  50         20000      1
398  15755018   Male   36         33000      0
399  15594041  Female  49         36000      1

[400 rows x 5 columns]>
```

```
In [7]: df.drop(columns=["User ID"],inplace=True)
df['Gender']=df['Gender'].map({'Male':1,'Female':0})
df
```

Out[7]:

	Gender	Age	EstimatedSalary	Purchased
0	1	19	19000	0
1	1	35	20000	0
2	0	26	43000	0
3	0	27	57000	0
4	1	19	76000	0
...	...	...	...	...
395	0	46	41000	1
396	1	51	23000	1
397	0	50	20000	1
398	1	36	33000	0
399	0	49	36000	1

400 rows × 4 columns

```
In [8]: x = df[['Gender', 'Age', 'EstimatedSalary']]
        y = df['Purchased']
```

```
In [12]: x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.20, random
```

```
In [13]: scaler = StandardScaler()
        x_train = scaler.fit_transform(x_train)
        x_test = scaler.transform(x_test)
```

```
In [15]: model = LogisticRegression()
        model.fit(x_train, y_train)
```

```
Out[15]: LogisticRegression
         LogisticRegression()
```

```
In [16]: y_pred = model.predict(x_test)
```

```
In [17]: accuracy = accuracy_score(y_test, y_pred)
        accuracy
```

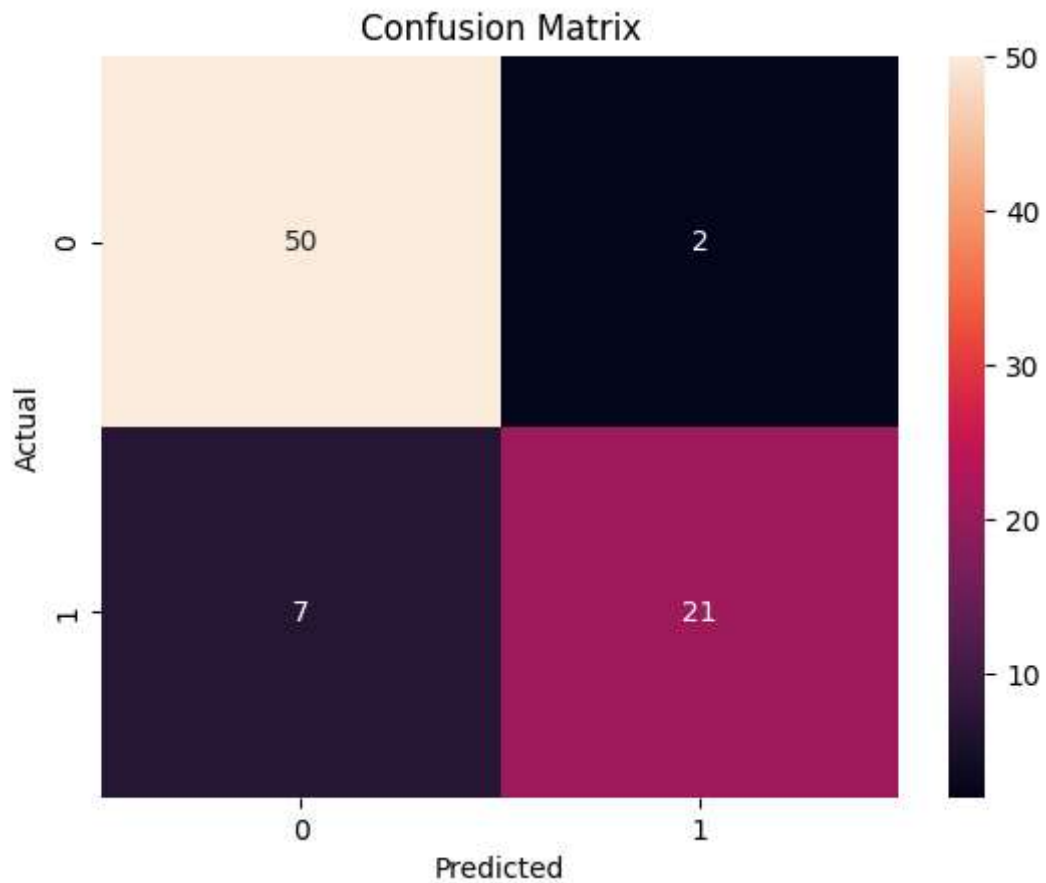
Out[17]: 0.8875

```
In [18]: conf_matrix = confusion_matrix(y_test, y_pred)
        conf_matrix
```

```
Out[18]: array([[50,  2],
               [ 7, 21]])
```

```
In [19]: sns.heatmap(conf_matrix, annot=True)
        plt.xlabel("Predicted")
```

```
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```



```
In [21]: new_data = np.array([[1,30,80000]])
new_data_df = pd.DataFrame(new_data , columns = x.columns)
new_data_scaled = scaler.transform(new_data_df)
prediction = model.predict(new_data_scaled)
if prediction == 1:
    print("Purchase Prediction: Yes")
else:
    print("Purchase Prediction : No")
```

Purchase Prediction : No